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Title of the article: Confidence reports in decision-making with multiple alternatives violate the Bayesian confidence hypothesis

Extracted content:

While human observers can report their self-assessment of the quality of their decisions^{6 12}, the computations underlying confidence reports are still insufficiently understood. The leading theory of confidence suggested that confidence reflects the probability that a decision is correct^{7,8,13 17}.

We refer to this idea as the Bayesian confidence hypothesis, meaning that the decision-makers use the posterior probability of the chosen category (i.e. the subjective probability that decision is correct) for their confidence reports.

Accordingly, in neurophysiological studies, a brain region or a neural process is considered to represent confidence if its responses correlate with the probability that a decision is correct^{18 20}.

Even though the Bayesian confidence hypothesis is the leading theory of confidence, there is currently no evidence to rule out the possibility that confidence is affected by the probability of correctness of the unchosen options.

Specifically, people could be less confident if the next-best option is very close to the best option. In other words, confidence could depend on the difference between the posterior probabilities of the best and the next-best options, rather than on the absolute value of the posterior of the best option. The reason that this idea has not been tested before might be that previous studies of decision confidence predominantly used two-alternative decision tasks; in such tasks, the alternative hypothesis is equivalent to the Bayesian confidence hypothesis, because the difference between the two posterior probabilities in a two-alternative task is a monotonic function of the highest posterior probability.

To preview our main result, we find that the difference-based model accounts well for the data, whereas the model corresponding to the Bayesian confidence hypothesis and a third, entropy-based model do not.

This design allowed us to test quantitative models of how the posterior distribution gives rise to confidence reports (see an illustration of this idea in Supplementary Fig. 1).

Given a measurement x , the posterior probability of category C is then $p(C | x)$. We further assume that due to decision noise or inference noise, the observer does not maintain the exact posterior distribution, $p(C | x)$, but instead a

noisy version of it.

We modeled decision noise by drawing a noisy posterior distribution from a Dirichlet distribution around the true posterior (Fig. 2a, b; See details in Methods).

Unless otherwise specified, we will from now on refer to the noisy posterior distribution as simply the posterior distribution. We introduce three models of confidence reports: the Max model, the Entropy model and the Difference model.

Each of these models contains two steps: (a) mapping the posterior distribution (q) to a real-valued internal confidence variable; (b) applying three criteria to this confidence variable to divide its space into four regions, which, when ordered, map to the four confidence ratings.

In this model, the confidence variable is the probability that the chosen category is correct, or in other words, it is the highest of the three posterior probabilities (Fig. 2c).

Importantly, after the posterior distribution is computed, the posterior probability of the unchosen options does not further contribute to the computation of confidence.

In the Difference model, the confidence variable is the difference between the highest and second-highest posterior probabilities.

One interpretation of this model is that confidence reflects the observer's subjective probability that they made the best possible choice, regardless of the actual posterior probability of that choice.

In the Entropy model, the confidence variable is the negative of the uncertainty conveyed by the entire posterior distribution, as quantified by its negative entropy.

Unlike in the Max model, however, the posterior probabilities of the non-chosen categories directly affect confidence.

Thus, in all three models, the unchosen options implicitly affect confidence by contributing to the denominator in the computation of the posterior probabilities.

The three models differ in how the confidence variable is read out from the posterior distribution. The Max model is unique in assuming that after the computation of the posterior probabilities, the unchosen categories do not further affect the computation of confidence.

In our three-alternative task, these models generate qualitatively different mappings from the posterior distribution to the confidence variable (Fig. 2c).

Because the Max, the Difference and the Entropy used the same rule for category decisions, we compared

category decision models that used the same decision rule (reporting the category with the highest posterior probability), but included sensory noise only, decision noise only, or both.

We found that the Difference model outperformed the Max and the Entropy models when we replaced Dirichlet decision noise by drawing samples from the true posterior, or when we added noise in the measurement of the category means (Supplementary Fig. 10A).

In addition, we tested heuristic models that made category decisions and confidence reports based on the category means and the noisy measurement of the target location (x) but did not compute posterior probabilities.

We sampled the target dot positions uniformly within a circular area centered on the screen. In addition, the distribution of the categories used in Experiment 2 allowed us to probe confidence reports in a wider range of posterior distributions (Supplementary Fig. 1B).

However, whether participants report the probability that a decision is correct (the Max model) might depend on the experimental design. In Experiment 1 and 2, participants received no feedback on their category decision.

Thus, the probability of being correct in the task could be difficult to learn. To investigate this issue, in Experiment 3, using the same four stimulus configurations as those in Experiment 1 (Fig. 1b), we randomly chose one of the three groups as the true target category in each trial, and sampled the target position from the distribution of the true category.

Discussion To distinguish the leading model of perceptual confidence (the Bayesian confidence hypothesis) from a new alternative model in which confidence is affected by the posterior probabilities of unchosen options, we studied human confidence reports in a three-alternative perceptual decision task.

We found that confidence is best described by the Difference model, in which confidence reflects the difference between the strength of observers' belief (posterior probability) of the top two options in a decision. The Max model (which corresponds to the Bayesian confidence hypothesis) and the Entropy model (in which confidence is derived from the entropy of the posterior distribution) fell short in accounting for the data.

Our results demonstrate that the posterior probabilities of the unchosen categories impact confidence in decision-making.

This serves as a coarse approximation to the curvature of the posterior distribution at its peak, which, for Gaussians, is monotonically related to its inverse variance, consistent with an earlier model by van den Berg et al.²⁹, in which confidence is based on the precision parameter of the posterior in continuous estimation tasks²⁹.

For example, when the posterior distribution is bimodal, with the modes slightly different in height, the variance of the posterior is dominated by the separation between the modes, whereas the Difference model will

use the difference in height for confidence reports.

In addition to the probability of being correct, confidence is influenced by various factors such as reaction time³², post-decision processing^{33 36}, and the magnitude of positive evidence^{37 40}.

First, the Difference model might be an approximation to a model in which confidence depends on the probability that an observer made the best possible decision. In this view, the observer possesses metacognitive knowledge that their decision is based on the noisy posterior q rather than the true posterior p , and consequently, realizes that it is possible that the chosen category is not the category with the highest true posterior probability. Confidence would then be derived from the probability that the chosen category has the highest true posterior probability.

Hence, an alternative interpretation of our results is that confidence reports deviate from the Bayesian confidence hypothesis (the Max model) because the observer estimates the probability of correct in a way that ignores the options that are discarded before final evaluation. In the Difference model, the least favorite option is not completely discarded because it decreases the posterior probabilities of the other two options (and thus their difference) by contributing to the normalization pool^{45,46}.

In a race model with two accumulators, they suggested that confidence is computed as the difference between the accumulated evidence of the two accumulators⁵¹.

In a two-alternative value-based decision task, observers reported confidence in a way that was similar to that in perceptual decision tasks:¹⁰ When observers were asked to choose the good with the higher value, confidence increased with the posterior probability that a decision is correct, which in turn increased with the difference in value between the two goods.

The Difference model would predict that, confidence is determined by the difference between the probability that the chosen item is the most valuable and the probability that the next-best item is the most valuable. Most neurophysiological studies of confidence have considered the neural activity that correlates with the probability of being correct as the neural representation of confidence (but see ref.

By utilizing multiple-alternative tasks, neural studies could dissociate the neural correlates of probability correct from that of the difference confidence variable in the Difference model, which according to our results, might be the basis of human subjective confidence.

A potentially important difference between human and non-human animal studies is that in the latter, confidence is not explicitly reported but operationalized through some aspect of behavior, such as the probability of

choosing a safe (opt-out) option^{19,20,55,59,60}, or the time spent on waiting for reward¹⁸.